

Karyotypes of many species of the genus *Sceloporus* support the generalization that there are no morphologically recognizable sex chromosomes in lizards.

32. How many chromosomes are there in humans?

- (a) 36 (b) 46
(c) 56 (d) 26

U.P.P.C.S. (Pre) 1992

U.P.P.C.S. (Pre) 1991

Ans. (b)

In humans, each cell normally contains 23 pairs of chromosomes, a total of 46. Twenty-two of these pairs called autosomes which are the same in both males and females. The 23rd pair, the sex chromosomes differ between males and females. Females have two copies of the X chromosome (XX) while males have one X and one Y chromosome (XY).

33. The number of chromosome pairs, found in the cells of human body is –

- (a) 21 (b) 22
(c) 23 (d) 24

Uttarakhand P.C.S. (Pre) 2003

Ans. (c)

See the explanation of above question.

34. Assertion (A) : In human being, the females play a major role in determining the sex of the offspring.

Reason (R) : Women have two 'X' chromosomes.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not a correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

I.A.S. (Pre) 2000

Ans. (d)

The X chromosome is one of the two sex-determining chromosomes (allosomes) in many animal species, including mammals and is found in both males and females. Each person normally has one pair of sex chromosomes in each cell. Females have two X chromosomes (XX) whereas males have one X and one Y chromosome (XY). X of male and X of female give birth to a female child (44+XX) whereas Y of male and X of female give birth to a male child (44+XY). Thus, in human being, the males play a major role in determining the sex of the offspring.

35. Male's gene constituent is –

- (a) XX (b) XY

(c) X

(d) Y

43rd B.P.S.C. (Pre) 1999

Ans. (b)

See the explanation of above question.

36. Which combination of chromosomes in mankind give birth to a male child?

- (a) X of male and X of female
(b) X of male and Y of female
(c) Y of male and X of female
(d) Y of male and Y of female

M.P.P.C.S. (Pre) 2005

Ans. (c)

See the explanation of above question.

37. Which of the following professional(s) are more likely to run the risk of a permanent change in their cell's DNA?

1. Researchers using carbon 14 isotope
2. X-ray technician
3. Coal miner
4. Dyer and painter

Select the correct answer by using the codes given below :

Code :

- (a) 2 only (b) 1, 2 and 3
(c) 1, 2 and 4 (d) 1, 3, and 4

I.A.S. (Pre) 1996

Ans. (a)

X-ray technicians are more likely to run the risk of a permanent change in their cell's DNA as compared to others.

Biofertilizers

Notes

- Over the years, chemical fertilizers have helped farmers increase crop production to meet the increasing demand. However, the use of these fertilizers is harmful because they cause air and water pollution; and also deplete minerals from the soil. Therefore, there is a need to now switch to organic farming which involves the use of biofertilizers.
- Biofertilizers increases the nutrients of host plants when applied to their seeds, plant surface or soil by colonizing the rhizosphere or the interior of the plant and promotes their growth. It refers to the use of microbes instead of chemicals to enhance the nutrition of the soil. As a result, it is also less harmful and does not cause pollution.

Types of Biofertilizers :

- Followings are the type of biofertilizers–

1. Bacteria
2. Fungi
3. Cyanobacteria

1. Bacteria:

- The great example of biofertilizers is the nitrogen-fixing nodules on the roots of legumes. The nodules are formed by the association of the bacterium '**Rhizobium**' with the roots of these plants. This association is beneficial and is, therefore, called **symbiotic**.
- The nodules help in fixing atmospheric nitrogen into organic forms which can then be used as nutrition by plants. Adding **Rhizobium culture** to fields has become a common practice to ensure an adequate amount of nitrogen in the soil.
- **Leghaemoglobin** is the haeme protein found in root nodules of leguminous plants, where they are crucial for supplying sufficient oxygen to root nodule bacteria for nitrogen fixation to occur. It is produced by legumes in response to the roots being colonized by nitrogen-fixing bacteria.
- Phosphate-solubilizing bacteria, such as **Pantoea agglomerans** (P5 strain), **Microbacterium laevaniformans** (P7 strain) and **Pseudomonas putida** (P13 strains), are able to solubilize the insoluble phosphate from organic and inorganic phosphate sources.
- Other examples of bacteria that act as biofertilizers include **Azospirillum** and **Azotobacter**. These bacteria are free-living in the soil. **Azotobacter** is usually used with crops like cotton, wheat, mustard, maize, etc. **Azospirillum** inoculations are recommended mainly for sugarcane, wheat, millets, maize and sorghum.

2. Fungi :

- Symbiotic association exist between plants and fungi too. These associations are called **Micorrhizae**. In this association the plant makes organic molecules such as sugars by photosynthesis and supplies them to the fungus, and the fungus supplies to the plant water and mineral nutrients, such as phosphorus, taken from the soil. Plants that grow with these associations also show other advantageous characteristics such as—
 1. Tolerance to drought conditions and salinity.
 2. Resistance to root borne pathogens.
 3. An overall increase in plant growth and development.
- #### 3. Cyanobacteria/Blue-Green Algae :
- These are blue-green bacteria found in water and on land. They also help in fixing atmospheric nitrogen. Examples are **Oscillatoria**, **Nostoc**, **Anabaena** etc.
 - Some cyanobacteria can fix atmospheric nitrogen in anaerobic conditions by means of specialized cells

called heterocysts. Heterocysts may also form under the appropriate environmental conditions (anoxic) when fixed nitrogen is scarce.

- The symbiotic association between the aquatic fern **Azolla** and **Anabaena** is very important for rice fields. In this association **Azolla** provides food to **Anabaena** and **Anabaena** provides N_2 to **Azolla** by fixing nitrogen.

Question Bank

1. Which of the following is/are used as biofertilizers?

1. **Azolla**
2. **Blue-green algae**
3. **Alfalfa**

Choose the correct answer by the given codes :

- | | |
|-------------|----------------|
| (a) Only 2 | (b) 1 and 2 |
| (c) 1 and 3 | (d) 1, 2 and 3 |

I.A.S. (Pre) 1994

Ans. (b)

Biofertilizers are defined as preparations containing living cells or latent cells of efficient strains of microorganisms that help crop plants uptake of nutrients, by their interactions in the rhizosphere when applied through seed or soil. They accelerate certain microbial processes in the soil which augment the extent of availability of nutrients in a form easily assimilated by plants. They can be grouped in different ways based on their nature and function.

Nitrogen-fixing Biofertilizers : *Azotobacter*, *Beijerinckia*, *Clostridium*, *Klebsiella*, *Anabaena*, *Nostoc*, *Rhizobium*, *Frankia*, *Azolla pinnata*, *Cyanobacteria* (Blue-green algae) etc.

Phosphorus-solubilizing/mobilizing Biofertilizers : *Bacillus megaterium* var. *phosphaticum*, *Aspergillus*, *Pezizella ericae* etc.

Biofertilizers for micronutrients : *Bacillus* spp.

Plant Growth Promoting Rhizobacteria : *Pseudomonas fluorescens* etc.

Cyanobacteria (blue-green algae) and *Azolla* are used as biofertilizers. However, *Alfalfa* is a perennial flowering plant of the pea family.

2. Which of the following is/are used as biofertilizers?

Select the correct answer from the codes 2 given below:

1. **Azolla**
2. **Blue-green algae**
3. **Alfalfa**
4. **Nitrolin**

Codes :

- | | |
|----------------|----------------|
| (a) 2 and 4 | (b) 1, 2 and 4 |
| (c) 2, 3 and 4 | (d) 1, 2 and 3 |

U.P.P.C.S. (Pre) 2002

U.P.P.C.S. (Pre) 2001

Ans. (*)

See the explanation of above question.

3. Which of the following is used as Biofertilizer?

- (a) Rhizobium (b) Blue-green algae
(c) Azolla (d) All of these

Chhattisgarh P.C.S. (Pre) 2019

Ans. (d)

Rhizobium, Blue-green algae (cyanobacteria) and Azolla, all are used as biofertilizer. They help in nitrogen-fixation.

4. Which of the following is not a nitrogen-fixing organism?

- (a) Nostoc (b) Anabaena
(c) Azotobacter (d) Mycorrhiza

U.P. P.C.S. (Mains) 2017

Ans. (d)

Nostoc, Anabaena and Azotobacter are nitrogen-fixing organisms while mycorrhiza is a symbiotic association between a fungus and a plant.

5. Which of the following is a source of bio-fertilizer?

- (a) Yeast (b) Chlorella
(c) Azolla (d) Mold

U.P.P.C.S. (Mains) 2009

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (c)

Azolla has the capability of nitrogen-fixing. That is why it is being widely used as a biofertilizer, especially in parts of Southeast Asia.

6. Which of the following is a bio-fertilizer ?

- (a) Urea (b) 2, 4 - D
(c) Streptocycline (d) Azolla

Uttarakhand P.C.S. (Mains) 2002

Ans. (d)

See the explanation of above question.

7. The biofertilizer used as a supplementary food for the cattle, especially for milking bovines is :

- (a) Azospirillum (b) Azolla
(c) Azotobacter (d) Rhizobium

R.A.S./R.T.S. (Pre) 2016

Ans. (b)

The demand for milk and meat in India is creating a new potential, in profitability of animal husbandry as an occupation. Yet, at some time there is a substantial decline in fodder availability. The search for alternatives led to a wonderful plant Azolla which holds the promise of providing a sustainable food for livestock. Azolla, in turn, provides the carbon source and favourable environment for the growth and development of a Algae. It is this unique symbiotic relationship that makes Azolla a wonderful plant with a high protein content.

8. Which of these is not correctly matched–

- (a) Blue-green algae – Bio-fertilizer
(b) Chrysopa – Maho-control
(c) Bio 902 – Variety of mustard
(d) Azotobacter – Bioinsecticide

U.P.Lower Sub. (Spl.) (Pre) 2003

U.P.Lower Sub. (Spl.) (Pre) 2002

Ans. (d)

Azotobacter is a bacteria. Azotobacter naturally fixes atmospheric nitrogen in the rhizosphere (the zone surrounding the root of plants). So option (d) is not correctly matched. Other three options are correctly matched.

9. Which of the following is associated with biological nitrogen fixation?

- (a) Red algae (b) Brown Algae
(c) Green algae (d) Blue-green algae

U.P.P.C.S. (Pre) 2019

Ans. (d)

Blue-green algae is also called Cynobacteria. It is associated with biological nitrogen fixation.

10. Blue-green algae are chiefly used as biofertilizer in the crop of –

- (a) Wheat (b) Gram
(c) Paddy (d) Mustard

U.P.P.C.S. (Pre) 2006

Ans. (c)

Cyanobacteria or blue-green algae is an example of biofertilizer, which helps in fixing atmospheric nitrogen. Blue-green algae are mainly used as biofertilizer in the crop paddy.

11. Blue-green algae is used to supply nitrogen to –

- (a) Bajra (b) Maize
(c) Paddy (d) Wheat

U.P.P.C.S. (Mains) 2010

Ans. (c)

See the explanation of above question.

12. Which one of the following is not correctly matched?

Nitrogen fixing agent **Crop**

- (a) Blue green algae – Paddy
(b) Rhizobium – Pea
Leguminisarum
(c) Azotobacter – Wheat
(d) Azolla – Maize

U.P.P.C.S. (Pre) 2017

Ans. (d)

Azolla, anabaena, nostok, blue-green algae etc. promote nitrogen fixation in paddy harvesting. They work as organic fertilizers. Rhizobium leguminisarum in pea and azotobacter wheat promotes nitrogen fixation. Generally azolla is not used for nitrogen fixation in maize. Thus the correct answer is option (d).

13. Azolla gives a good biofertilizer when mixed with –

- (a) Blue-green algae (b) Bone meal
(c) Cow dung (d) Urea

Uttarakhand P.C.S. (Pre) 2007

Ans. (a)

Azolla is an aquatic fern. It gives a good biofertilizer when mixed with blue-green algae (BGA) or cyanobacteria and contributes in supplying nitrogen to rice fields. In addition, they also bring about directly or indirectly a number of changes in the physical, chemical and biological properties of the soil and soil-water interface in rice fields.

14. Which of the following microorganisms are used as biofertilizers?

- (a) Virus (b) Fungi
(c) Protozoa (d) Cyanobacteria

U.P. R.O./A.R.O. (Mains) 2016

U.P. P.C.S. (Mains) 2012

Ans. (d)

See the explanation of above question.

15. Azolla is often used as biofertilizer because it is associated with –

- (a) Blue-green algae (b) Red algae
(c) Fungus (d) Moss

U.P.P.C.S. (Mains) 2011

Ans. (a)

See the explanation of above question.

16. Azolla is –

- (a) Aquatic Fern (b) Aquatic algae
(c) Aquatic bacteria (d) None of these

U.P. U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (a)

See the explanation of above question.

17. Aquatic Fern, which is used as a biofertilizer, is

- (a) Salvinia (b) Azolla
(c) Marsilia (d) Pteridium

U.P.R.O./A.R.O. (Mains) 2014

Ans. (b)

See the explanation of above question.

18. The aquatic plant commonly used as bio-fertilizer in the water-logged rice field is :

- (a) Wolfia (b) Azolla
(c) Trapa (d) Lemna

U.P. P.C.S. (Pre) 2023

Ans. (b)

Azolla is a floating aquatic fern that grows in temperate climates that are ideal for paddy cultivation. It is commonly used as bio-fertilizer in the water-logged rice field. Azolla pinnata is high in nitrogen, making it a perfect biofertilizer.

19. Which of the following is phosphatic biofertilizer?

- (a) Rhizobium
(b) Blue-green algae
(c) Azolla
(d) Vascular Arbuscular Mycorrhiza

U.P. B.E.O. (Pre) 2019

Ans. (d)

Rhizobium, Blue-green algae (Cyanobacteria) and Azolla help in nitrogen fixation, while Vascular Arbuscular Mycorrhiza (VAM) helps in uptake of phosphorus by the plant and hence, it is a phosphatic bio-fertilizers. VAM fungi benefit their host plant by improving the uptake of water, minerals and particularly the poorly mobile ion phosphorus in the soil. It also increases the plant's resistance against pathogens and improves plant tolerance to environmental stresses such as drought.

20. What is the use of biochar in farming?

1. Biochar can be used as a part of the growing medium in vertical farming.
2. When biochar is a part of the growing medium, it promotes the growth of nitrogen-fixing microorganisms.
3. When biochar is a part of the growing medium, it enables the growing medium to retain water for longer time.

Which of the statements given above is / are correct?

- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2020

Ans. (d)

Biochar is a stable, carbon-rich form of charcoal that is applied to the soil. It is produced from heating organic materials like crop waste, grass, woodchips and manure in a high temperature, low oxygen process known as pyrolysis. Biochar can be used as a part of the growing medium in vertical farming. Adding biochar to soil increases its carbon content and can mitigate greenhouse gas emissions. Some

biochars can improve soil fertility, water holding capacity and crop productivity. Research shows that biochars derived from grasses or crops appear to have the best balance of agricultural benefit and carbon stability. Some studies also show that increasing the application rate of biochar has a positive feedback on biological nitrogen fixation (BNF) and soil microbial biomass. The extremely porous nature of biochar is found to be effective at retaining both water and water-soluble nutrients, and it is very much suitable as a habitat for many beneficial soil microorganisms. From the above explanation, all of the given statements are correct.

21. Which of the following statements is incorrect in reference to compost?

- (a) In this, biodegradable waste is decomposed in pits.
- (b) It is rich in organic matter and nutrients.
- (c) The process of decomposition in pits is slowed down by earthworms.
- (d) The waste material decomposed is farm waste material like livestock excreta, vegetable waste etc.

R.A.S./R.T.S. (Pre) 2018

Ans. (c)

In the making of compost, the process of decomposition in pits is accelerated by earthworms and other microbes like bacteria and fungi. Hence, statement of option (c) is incorrect, while other three statements are correct.

22. The capability of nitrogen use efficiency can be increased in crop production by –

- (a) Use of split application
- (b) Use of nitrogen inhibitors
- (c) Using the slow release of nitrogen fertilizer
- (d) All of the above.

U.P.P.C.S. (Pre) 2009

Ans. (d)

The capability of nitrogen use efficiency can be increased by all of the given options; frequent use of fertilizer in the split application, use of nitrogen inhibitors and by using the slow release of nitrogen fertilizer. There are various bacteria in various crops which are liable for nitrogen stabilization.

23. Nitrogen fixing bacteria make combination with cells of the roots of :

- (a) Pulses
- (b) Rice
- (c) Wheat
- (d) Sugarcane

U.P.P.C.S. (Pre) 2019

Ans. (a)

Generally, the nitrogen-fixing bacteria are found in the roots of legumes (plants which give pulses) in special structure called root-nodules.

24. Which of the following are nitrogen-fixing plants?

- 1. Alfalfa
- 2. Amaranth
- 3. Chickpea
- 4. Clover
- 5. Purslane (Kulfa)
- 6. Spinach

Select the correct answer using the code given below:

- (a) 1, 3 and 4 only
- (b) 1, 3, 5 and 6 only
- (c) 2, 4, 5 and 6 only
- (d) 1, 2, 4, 5 and 6

I.A.S. (Pre) 2022

Ans. (a)

Plants that contribute to nitrogen fixation include the legume family – Fabaceae – with taxa such as clover, pigeon pea, chickpea, soyabean, alfalfa, lupinus, peanut, and rooibos, etc. They contain symbiotic bacteria called Rhizobia within nodules in their root systems, producing nitrogen compounds that help the plant to grow and compete with other plants. Amaranth, purslane (kulfa) and spinach are not nitrogen-fixing plants.

25. Which of the following plants is not helpful in nitrogen fixation?

- (a) Gram
- (b) Pea
- (c) Bean
- (d) Paddy
- (e) None of the above

Chhattisgarh P.C.S. (Pre) 2017

Ans. (d)

Legume crops such as gram, peas, beans, peanuts, soy, etc. are helpful in nitrogen fixation. Roots of legumes have nodules inside which Rhizobium bacteria are present. These bacteria take up nitrogen from the atmosphere and convert it into ammonia. Paddy fields generally lack nitrogen which is supplemented by the fertilizers.

26. Which bacteria is responsible for nitrogen fixation in soyabean?

- (a) Rhizobium leguminosarum
- (b) Rhizobium japonicum
- (c) Rhizobium fajiroleai
- (d) Rhizobium traitolai

U.P.P.C.S. (Pre) 2006

Ans. (b)

The Rhizobium japonicum bacteria is responsible for the nitrogen fixation in soyabean. Soyabeans inoculated with Rhizobium japonicum strains that synthesized the hydrogenase system fix significantly more nitrogen.

27. Leg-haemoglobin is found in–

- (a) Human blood
- (b) Rabbit blood
- (c) Legume root nodules
- (d) Chicken blood

U.P.P.C.S. (Mains) 2005

Ans. (c)